CADE User's Guide

Official Quick Reference & Operations Manual | Version 0.2.4

1. Welcome to CADE

CADE (Coding AI-Assistant with Desktop Extensions) is your stateful, local-first AI development environment. It operates as an intelligent engineering partner directly inside your terminal and web browser, automating code edits, terminal commands, and project management while maintaining maximum security, zero-panic stability, and elite token conservation.

Key System Highlights:

- **Cross-Platform Parity:** Engineered to run natively on Linux, macOS (Intel & Apple Silicon), and Windows 11.
- **Client-Daemon Architecture:** Uses a lightweight terminal client (`cade`) connected to a background HTTP daemon (`cade-server-bin`).
- **Stateful Memory & RAG:** Retains decision context across sessions using tiered memory blocks, hybrid vector search, and background compaction.

2. Quick Start & Terminal Execution

To launch CADE on any system, simply open your terminal and run:

```
$ cade
```

CADE automatically probes the background server. If offline, it auto-spawns `cade-server-bin`, performs health checks, and drops you directly into the interactive Terminal User Interface (TUI).

CLI Argument / Command	Action / Description
cade	Launches CADE interactive session (auto-boots background server daemon).
cade --check-update	Queries GitHub releases for new CADE updates and exits immediately.
cade --update	Downloads, cryptographically verifies (SHA256), and applies latest updates.
cade -p "<prompt>"	Runs a single-shot headless prompt without entering the interactive TUI.
cade --continue	Resumes the previous session immediately, suppressing env re-injection.

3. Dual-Interface System: TUI vs. Web Dashboard

Ratatui Terminal UI (TUI)	WebAssembly Dashboard (GUI)
• Fast, keyboard-driven alternate-screen TUI. • Ctrl+N: Start fresh conversation. • Ctrl+T: Toggle live plan checklist. • Ctrl+O: Toggle timeline expansion. • Tab: Cycle permission execution modes. • Type @ to trigger interactive file picker.	• Reactive WASM Web Dashboard (Dioxus 0.5). • Runs at http://localhost:8284/dashboard. • Real-time SSE event stream synchronization. • Instant 0ms Optimistic UI updates. • Interactive network topology graphs. • Full provider CRUD & approval queues.

4. Memory, RAG & Essential Slash Commands

CADE employs a **tiered memory system** backed by SQLite and hybrid RAG (FTS5 BM25 keyword ranking + `sqlite-vec` vector embeddings). As conversations expand, the background Sleeptime agent automatically condenses older turns into durable `session_summary` blocks.

Slash Command	Function & Usage
/help	Show the interactive menu of all available slash commands.
/memory	Inspect and edit persistent agent memory blocks (human, project, persona).
/compact	Manually trigger context compaction and archive older conversation turns.
/checkpoint (or /cp)	Create a working-tree snapshot (git commit) before risky operations.
/undo	Restore the codebase to the most recent working-tree checkpoint instantly.
/mcp	View, enable, or disable external Model Context Protocol (MCP) tool servers.
/skills	Load or browse domain knowledge packs (e.g. /skill:tdd, /skill:rust10x).

5. Subagents & Mixture of Agents (MoA)

CADE can delegate complex, long-running tasks (like codebase exploration, refactoring, or multi-file reviews) to parallel subagents. Subagents execute inside an isolated **Ephemeral Environment** and merge findings back into the primary workspace upon completion.

- **Single Delegation:** Run subagents using `/subagent `.
- **Team Mode:** Coordinate specialized teams (worker, reviewer, planner) using `run_team`.

6. Safety, Sandboxing & Enterprise Vector Storage

CADE prioritizes system safety with **water-tight directory canonicalization** and **permission execution modes**:

- **`plan` Mode:** Read-only execution. Prohibits file edits and shell mutations.
- **`acceptEdits` Mode:** Auto-applies file edits, but prompts for user approval prior to shell commands.
- **`yolo` Mode:** Full autonomous execution for experienced developers.

Enterprise Vector Adapters: CADE supports swapping its default local SQLite vector index with enterprise-scale **PostgreSQL (`pgvector`)** or **Qdrant** clusters using opt-in Cargo features (`--features enterprise-postgres` or `--features enterprise-qdrant`).